

Automating Thought of Search: A Journey Towards Soundness and Completeness

ICAPS 2026

Daniel Cao*
Cornell University

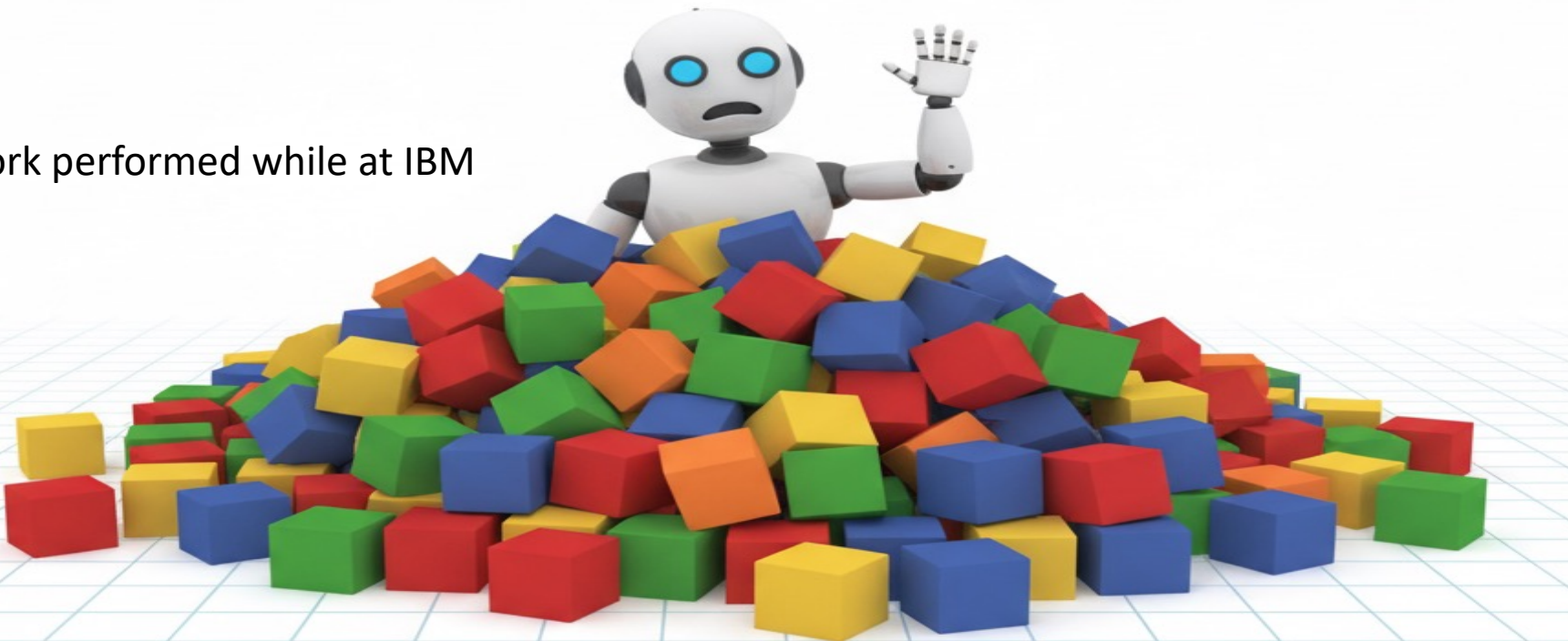
Michael Katz

Harsha Kokel
IBM

Kavitha Srinivas

Shirin Sohrabi

*Work performed while at IBM



Planning problem

Mathematical Model:

- finite and discrete state space S
- a **known initial state** $s_0 \in S$
- a set $S_G \subseteq S$ of goal states
- actions $A(s) \subseteq A$ applicable in each $s \in S$
- a **deterministic transition function**
$$s' = f(a, s) \text{ for } a \in A(s)$$
- non-negative **action costs** $c(a, s)$

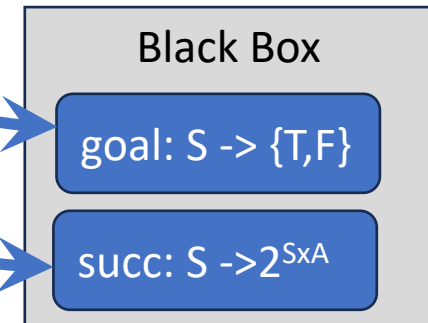
A **solution** is a sequence of applicable actions that maps s_0 into S_G , and it is **optimal** if it **minimizes sum of action costs**

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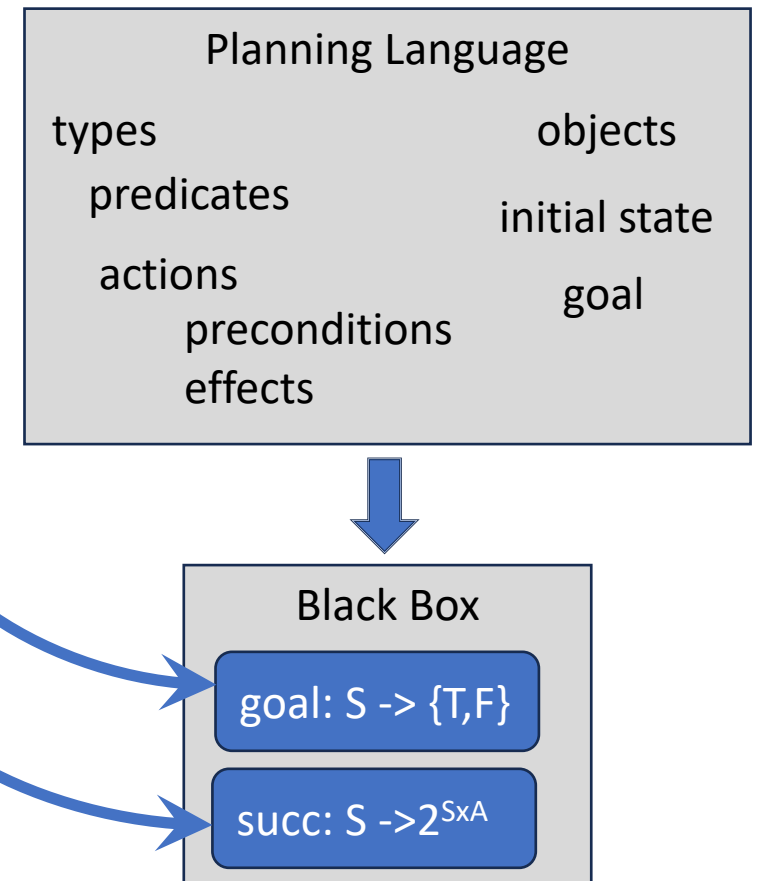


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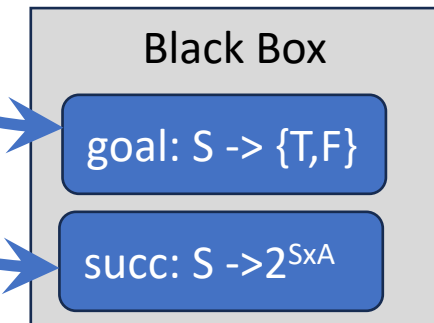
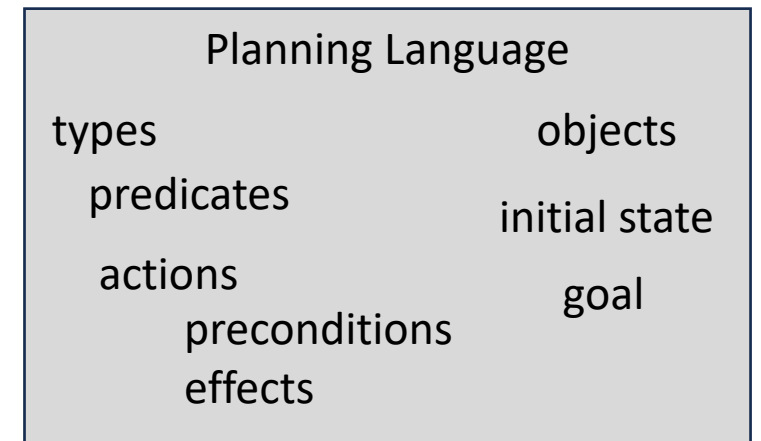
Planning problem

The 24 Game is a mathematical card game in which the objective is to find a way to manipulate four integers so that the end result is 24. The game is played with a list of four numbers, and the player must use all four numbers exactly once, using any combination of addition, subtraction, multiplication, or division, to arrive at the number 24.

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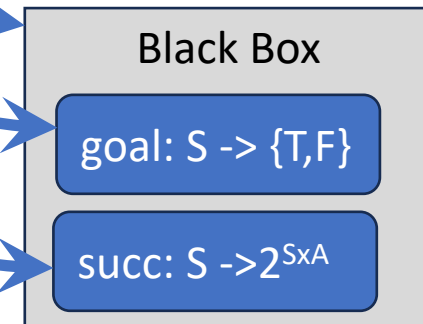
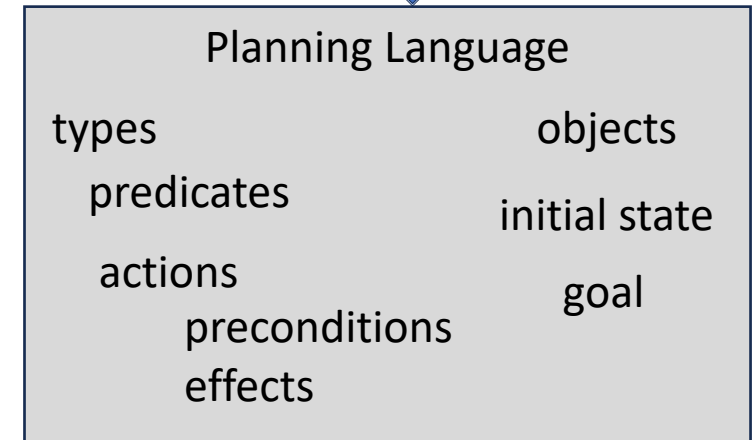
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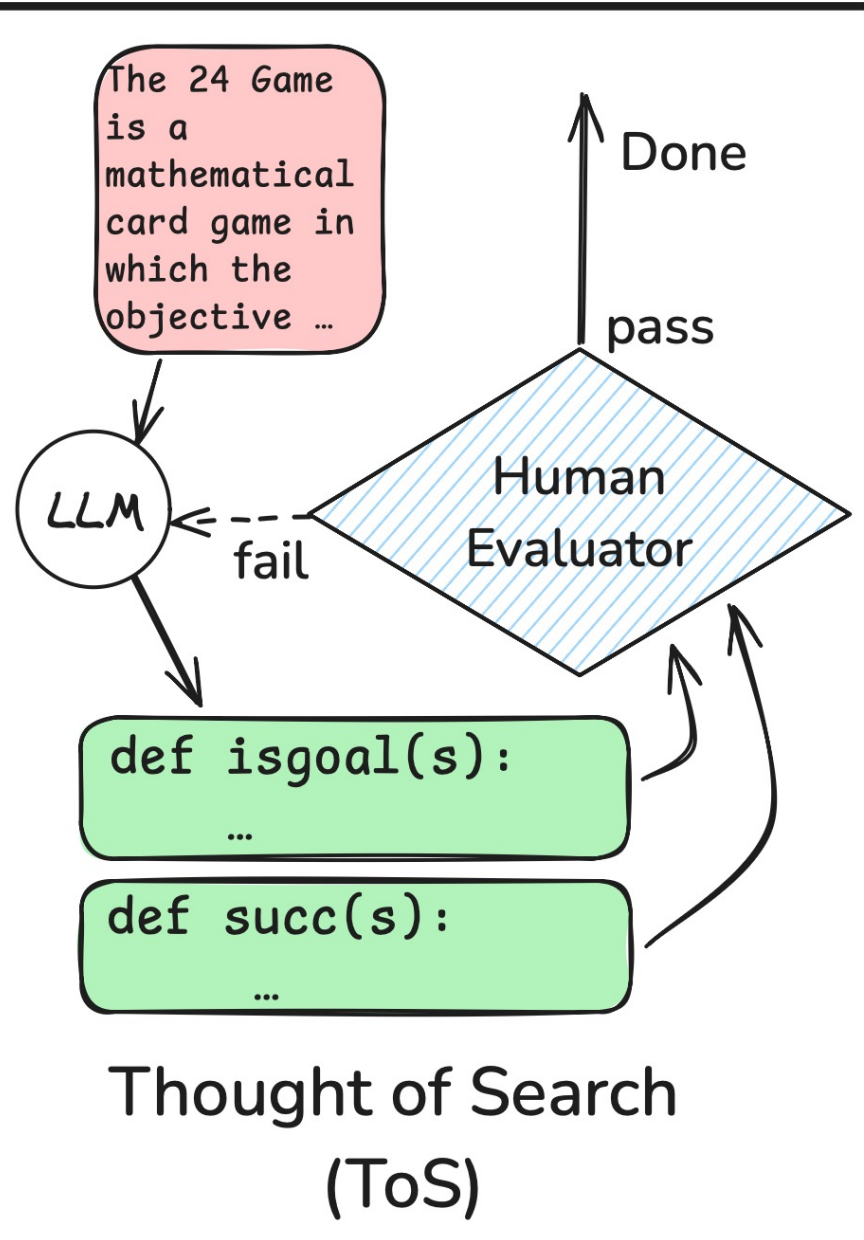
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Background: Thought of Search (ToS)

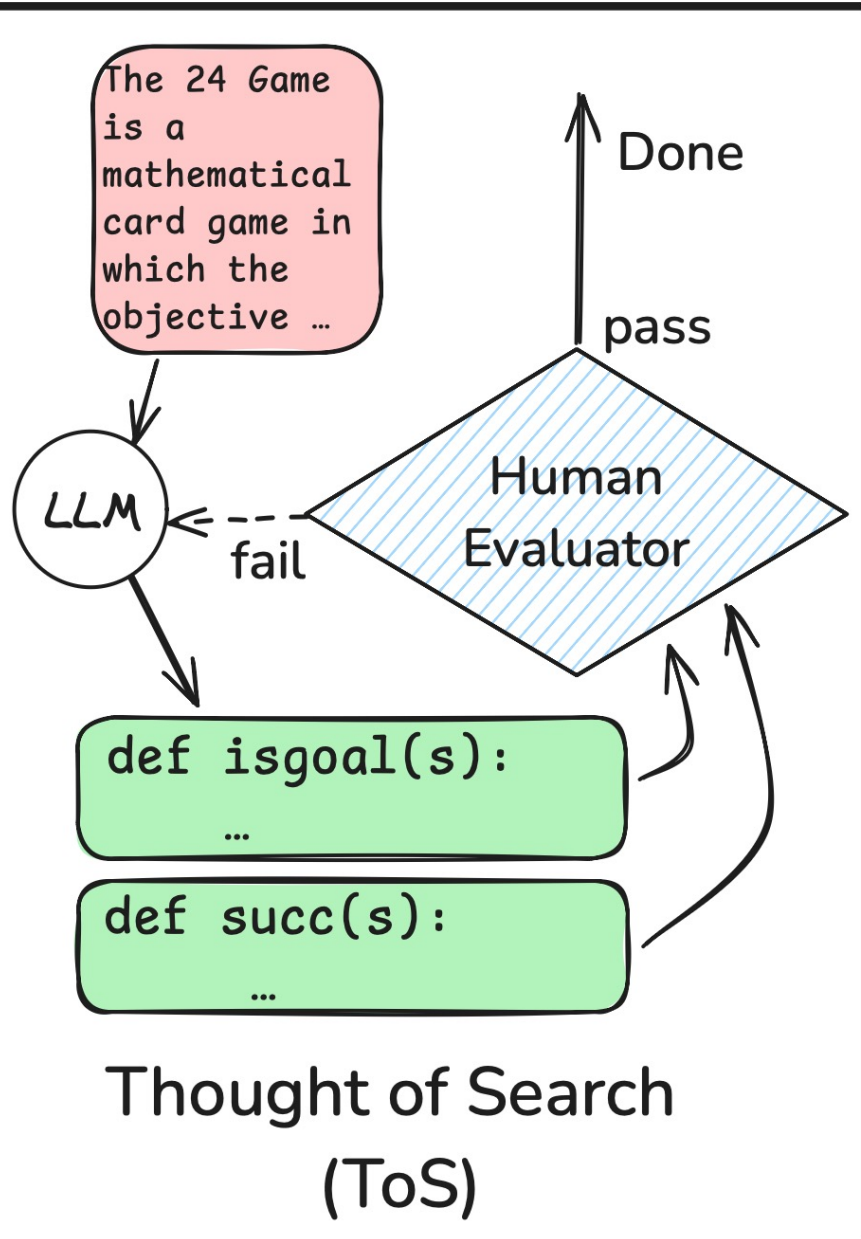
Katz et al, NeurIPS 2024



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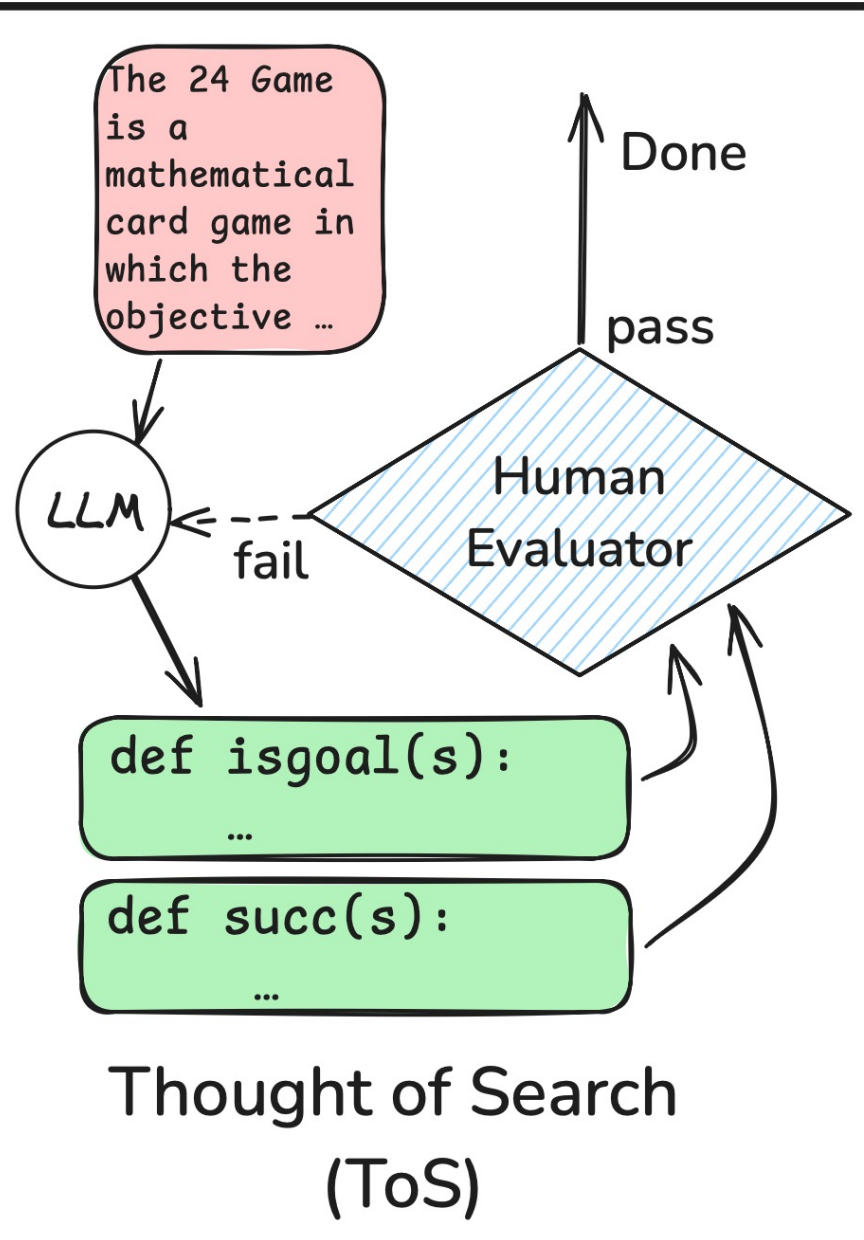
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Plug the functions into a standard search (BFS/DFS/IDDF...)



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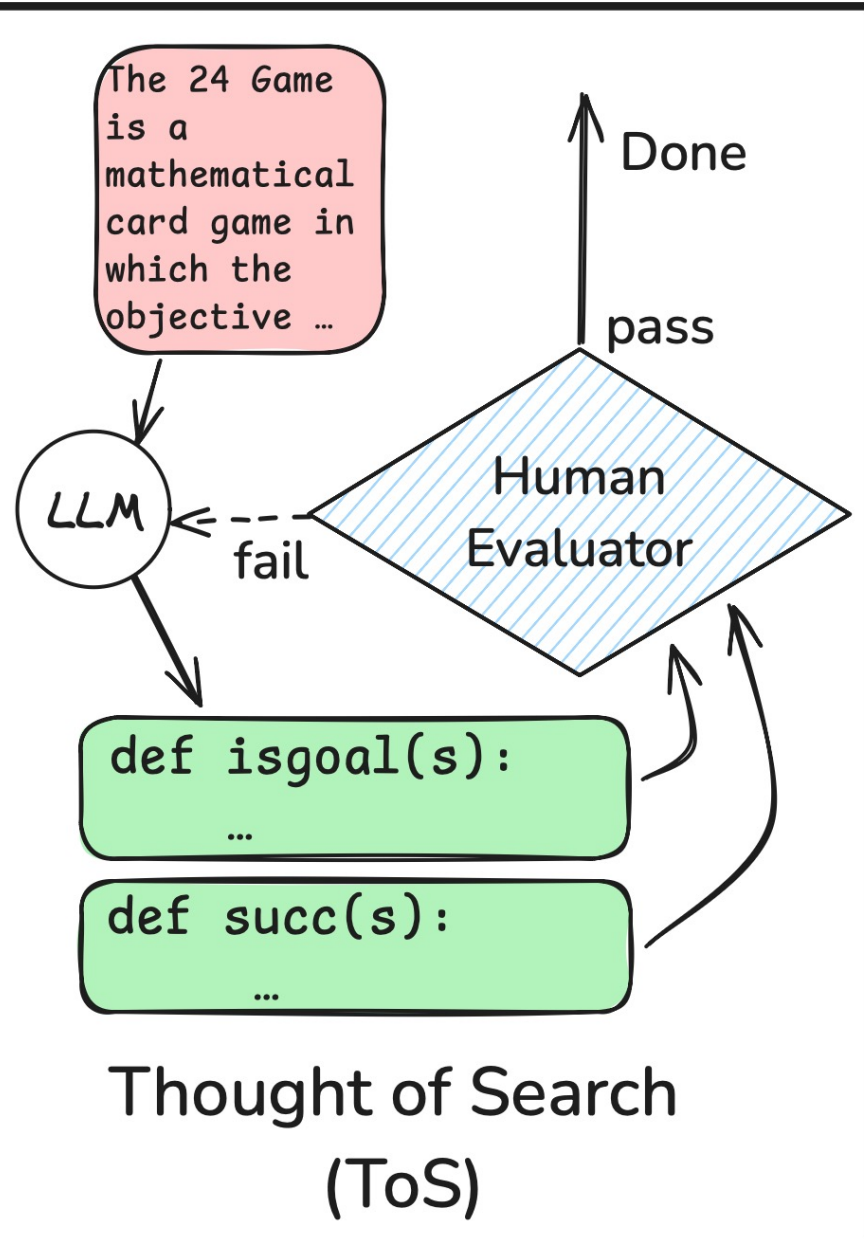
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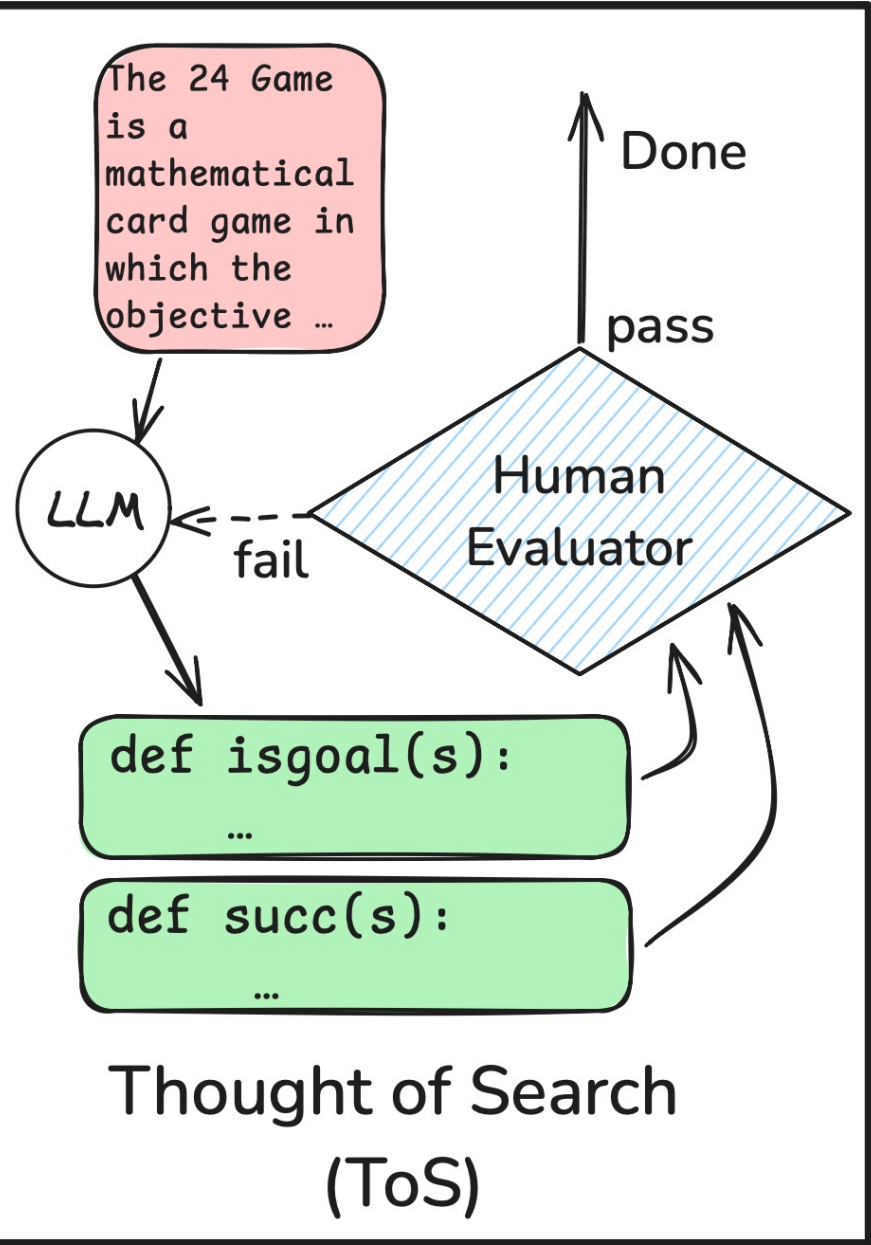
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Results?

- **100% accuracy across all tested domains!**
- **Execution time in ms per instance**

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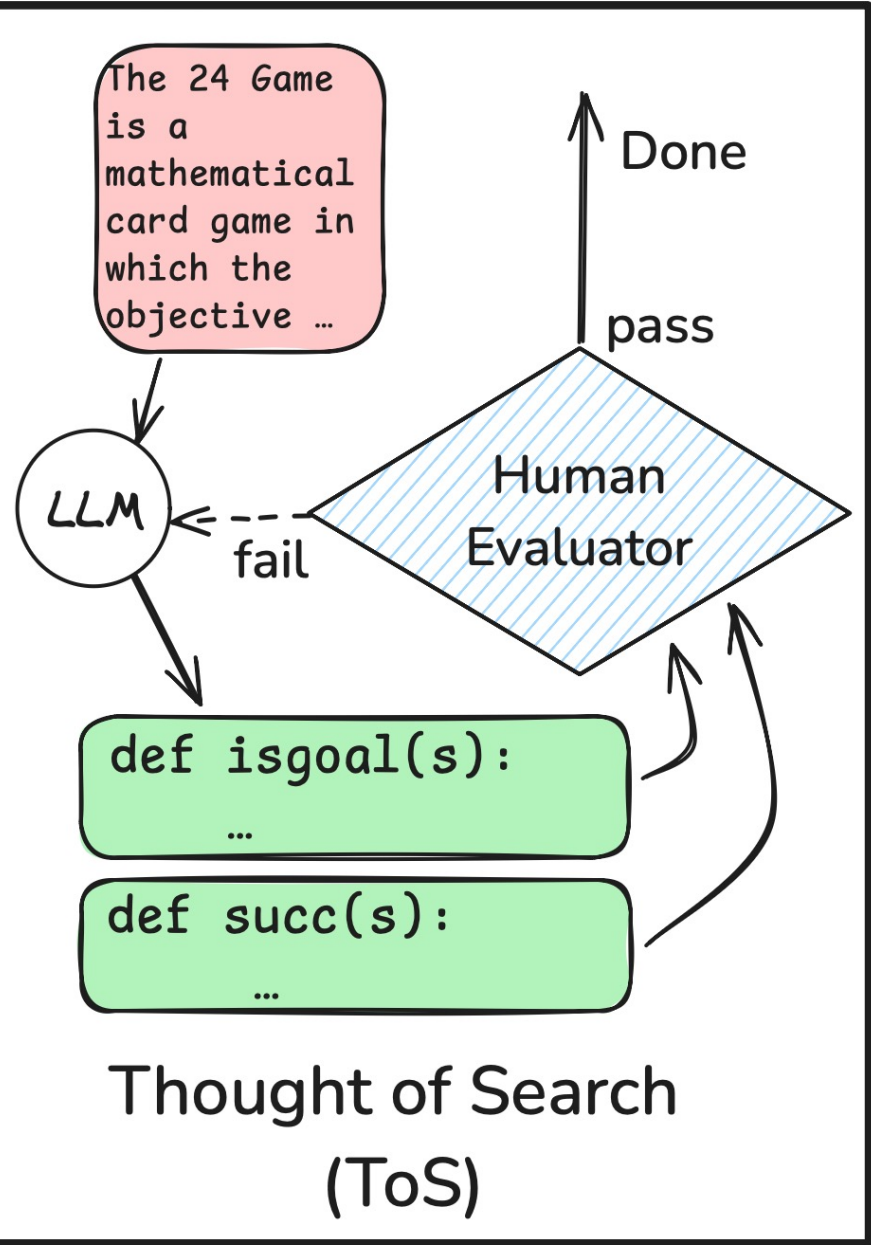
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		States	Calls	States	Calls	States	Calls	States	Calls
IO	O(D)	0.02%	1362	4e-9%	20	0.5%	502	4%	4000
CoT	O(D)	0.02%	1362	4e-9%	20	0.5%	502	4%	4000
ReAct	O(LD)	0.07%	4086	4e-8%	200	7.8%	8032	24.6%	24K
ReWOO	O(LD)	0.07%	4086	4e-8%	200	7.8%	8032	24.6%	24K
RAP	O(TbLD)	3.3%	245K	2e-6%	12K	388%	482K	1229%	1.44M
ToT	O(bmLD)	1.6%	102K	1e-6%	5K	194%	201K	615%	600K
GoT	O(bLD)	0.3%	20K	2e-7%	1K	39%	40K	122%	120K
Reflection	O(LTD)	0.7%	68K	4e-7%	2.4K	77.6%	90K	245%	320K
LATS	O(TbLD)	3.3%	286K	2e-6%	14K	388%	562K	1229%	1.68M
ToS (ours)	O(1)	27.0%	2.2	3e-4%	3.8	125%	3.8	175%	2.6

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Limitation: need human in the loop!

Automating Thought of Search (AutoToS)

Cao et al, ICAPS 2026

Issues Jobs

VentureBeat

Security

Data Infrastructure

Automation

Enterprise AI

AutoToS makes LLM planning fast, accurate and inexpensive



Image credit: VentureBeat with DALL-E 3



YouTube · Douglas Campbell

30+ views · 6 months ago



AutoToS: Revolutionizing AI Planning with Automated Search ...



Automated Logistics Planning: Use AutoToS to optimize supply chain and logistics operations, reducing the need for human planners. · Robotics ...



LinkedIn · David Borish

6 reactions · 6 months ago



AI That Fixes Itself: AutoToS and Self-Improving AI

AutoToS (Automated Thought of Search), a system that allows AI models to self-correct and improve their performance without human intervention.

About 18,000 results (0.18s)



MarkTechPost

<https://www.marktechpost.com> · 2024/08/26 · autotos-...

AutoToS: An Automated Feedback System for Generating ...

Aug 26, 2024 — AutoToS: An Automated Feedback System for Generating Sound and Complete Search Components in AI Planning.



Instagram

<https://www.instagram.com> · angelimarielim



AI Breakthrough Alert! IBM Research and Cornell University ...

IBM Research and Cornell University introduce AutoToS, a system that teaches AI to solve complex planning problems with 100% accuracy — no human supervision ...



newsbreak.com

<https://www.newsbreak.com> · news · 3574117154887-...

AutoToS: An Automated Feedback System for Generating Sound ...

AutoToS: An Automated Feedback System for Generating Sound and Complete Search Components in AI Planning. marktechpost.com. By Nikhil,. 2024-08-26.



Medium · EveryTechTrend

6 months ago



Automating Thought of Search: The Future of AI Planning with ...

We look at a new approach called AutoToS that automates the ToS process, allowing LLMs to generate high-performing search functions without any ...



Medium · Manoj Dadlani

6 months ago



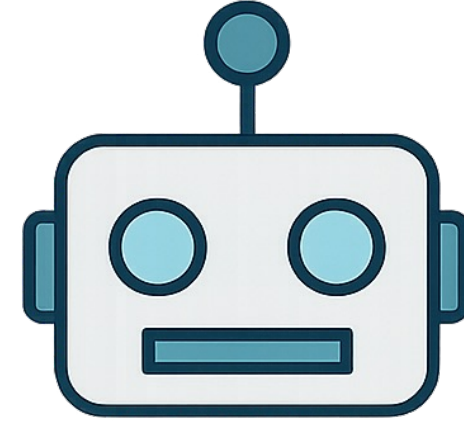
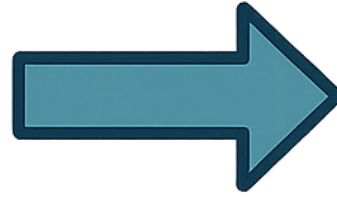
Automating AI Planning: How AutoToS is Revolutionizing ...

AutoToS operates by first generating an initial prompt that asks the LLM to produce a successor function and a goal test based on a natural ...

AutoToS: Core Idea



**Human
Evaluator**

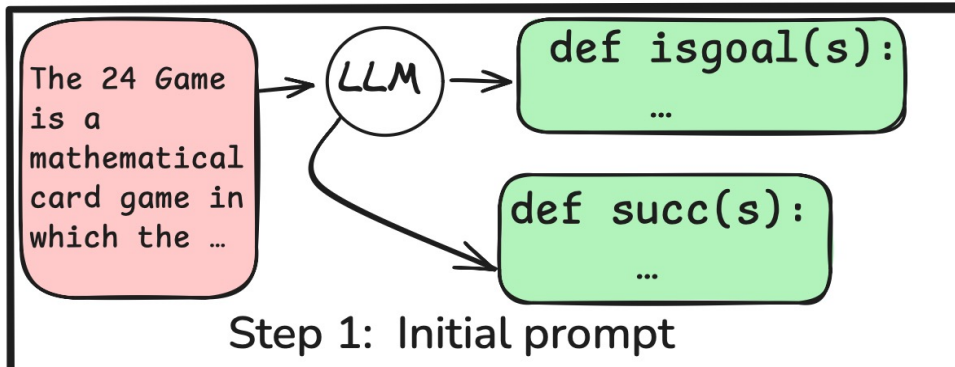


Generic and
domain-specific
unit tests

**Automated
Evaluator**

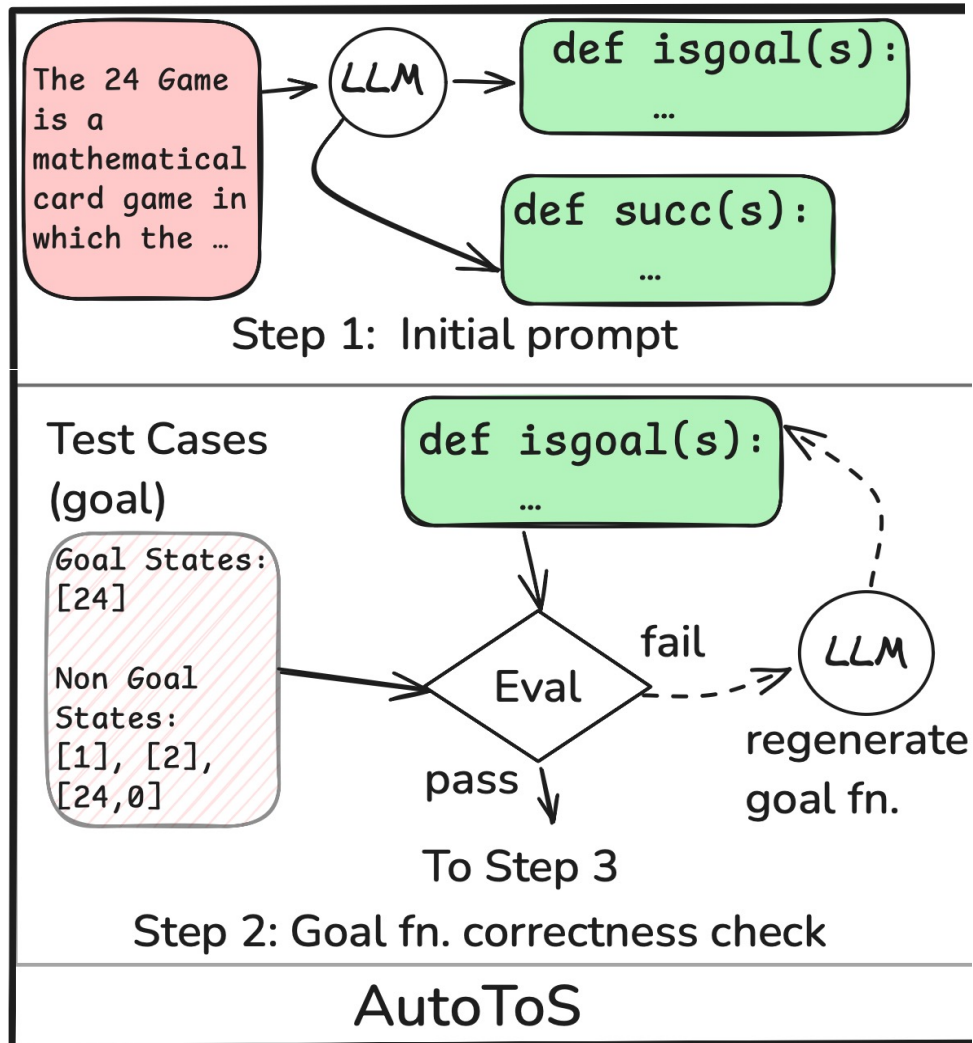
How does it work?

■ Problem Input ■ LM generated output ▨ Domain-specific test cases



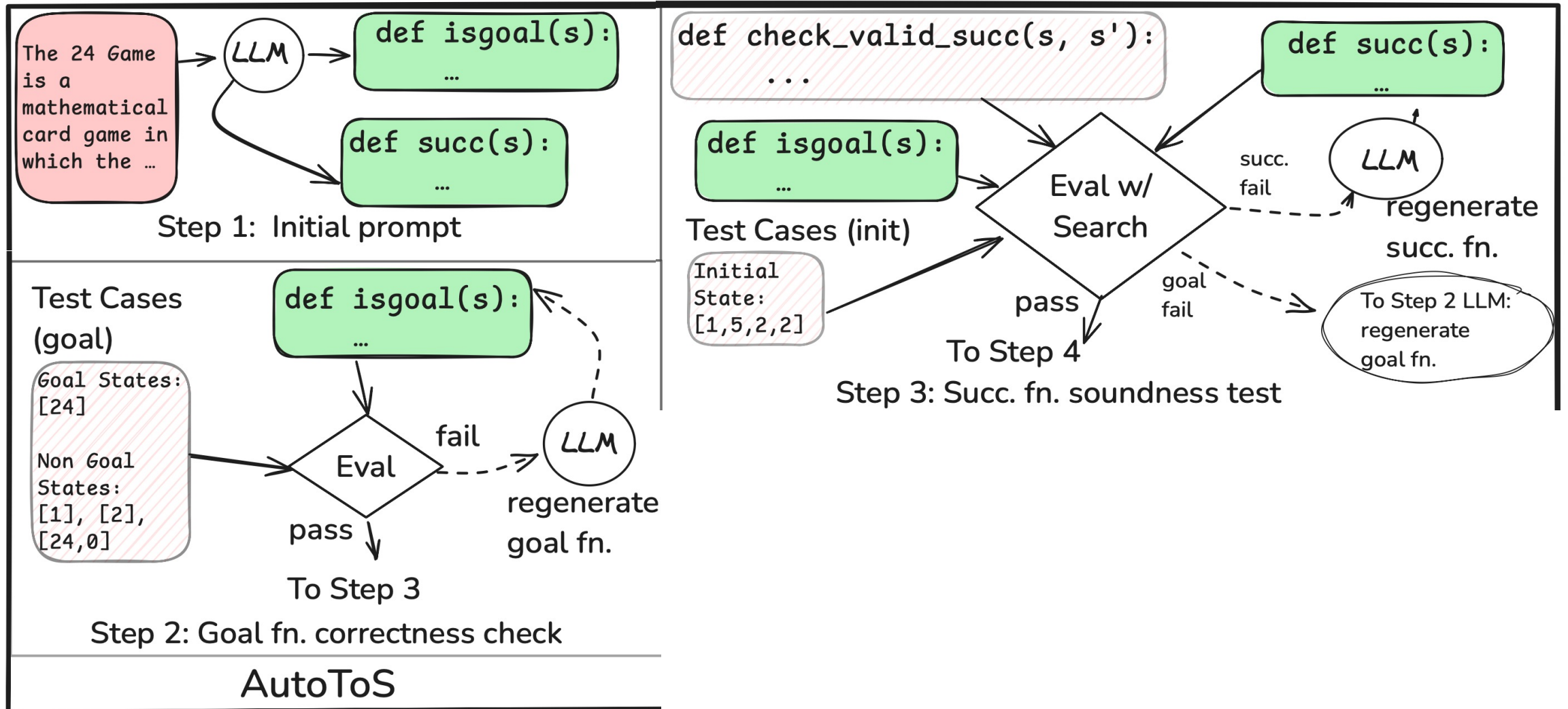
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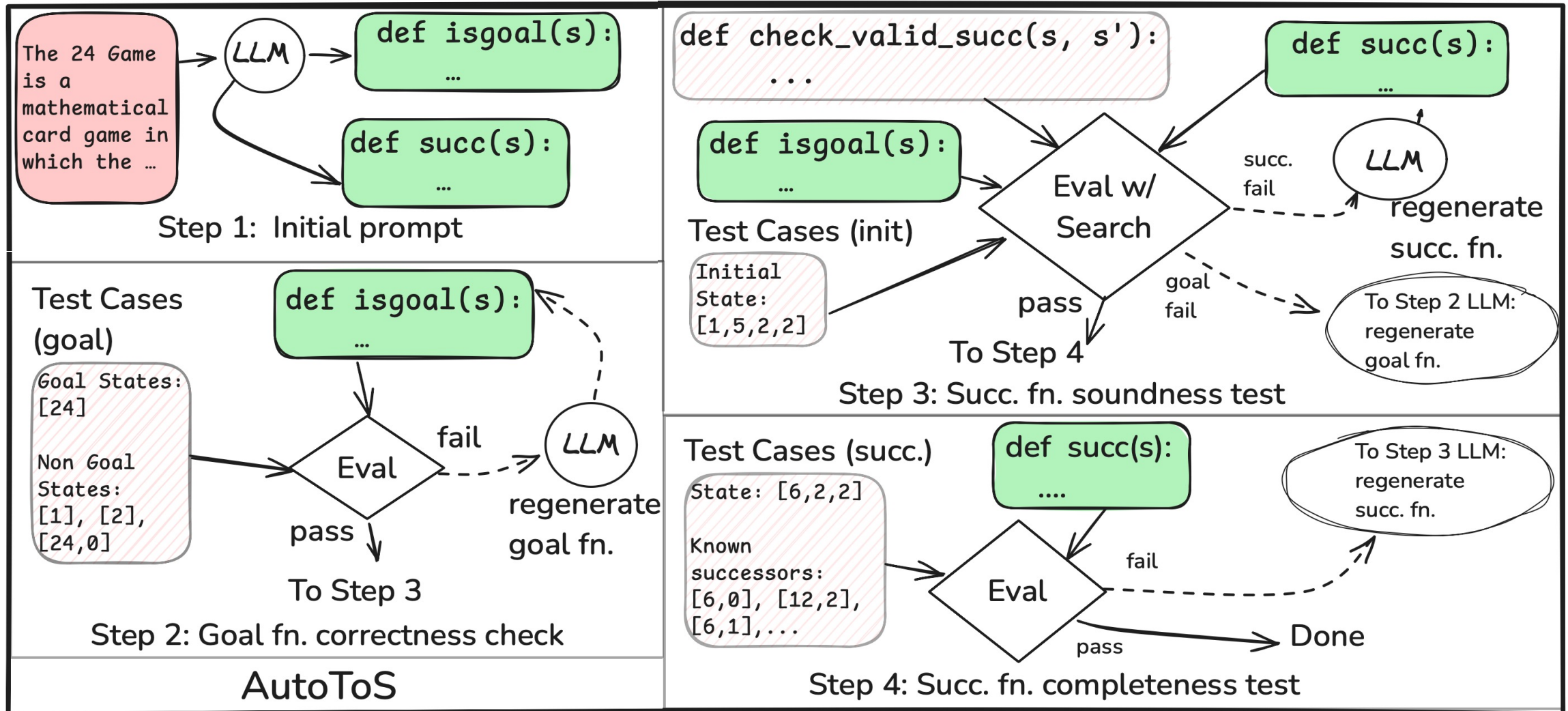
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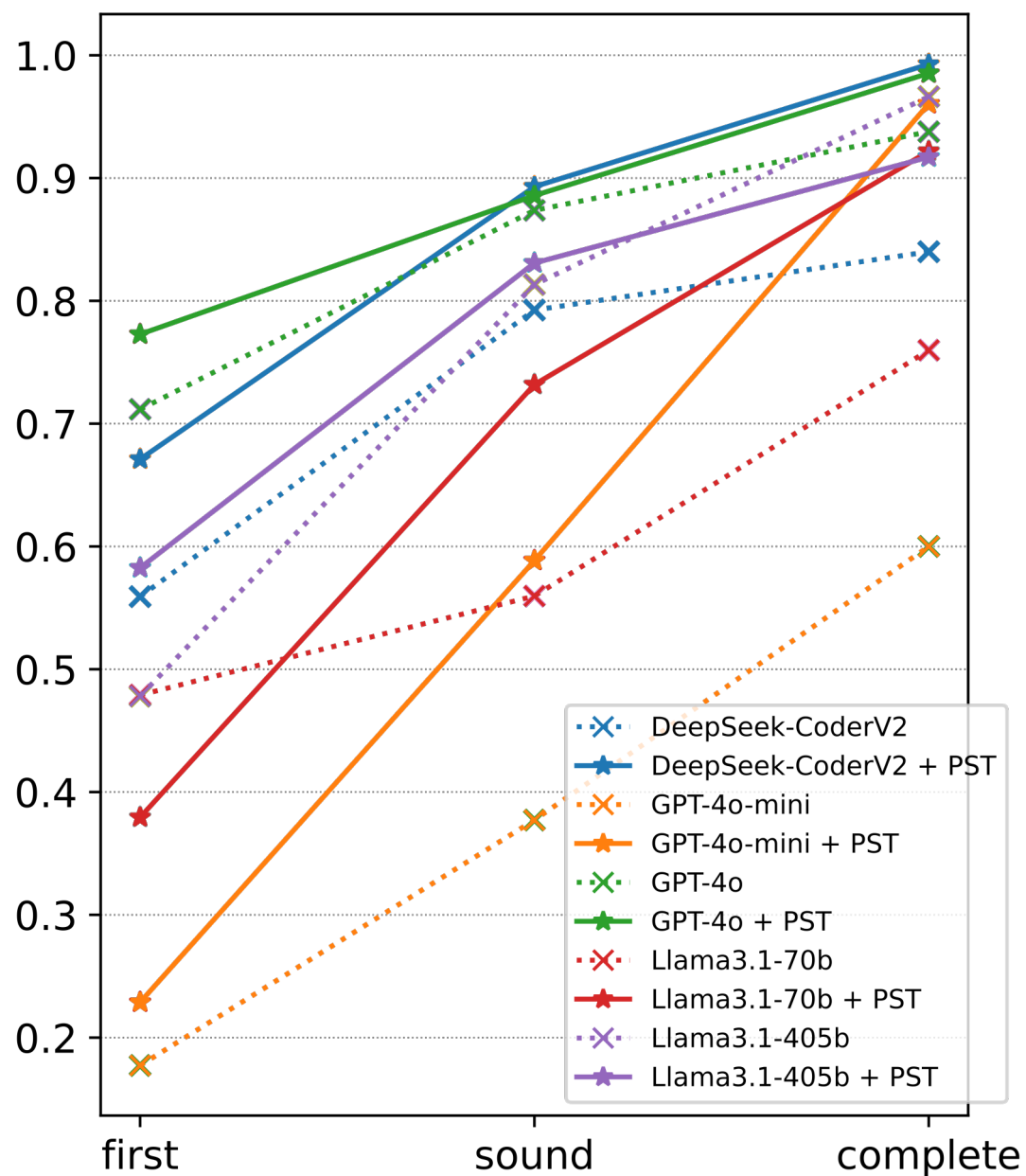
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Is it much worse than a human feedback?

		24 Game	PrOntoQA	Sokoban	Crossword	BlocksWorld
AutoToS	GPT-4o-mini	8.8	4.8	6.4	9.6	10.0
	GPT-4o	3.4	2.6	2.2	5.8	2.0
	Llama3.1-405b	3.4	2.0	2.6	4.0	3.2
	Llama3.1-70b	7.4	2.0	8.2	6.2	5.8
	DeepSeek-CoderV2	4.4	2.0	2.8	6.6	4.2
ToS	GPT-4	2.2	2.6	NA	3.8	3.8

Does it match the accuracy of ToS?



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	BlocksWorld			Sokoban		
model	F	S	C	F	S	C
DS-CoderV2+PST	1 (.53)	1 (1)	1 (1)	1 (.56)	1 (.76)	1 (.96)
GPT-4o-mini+PST	0 (0)	.83 (.21)	.83 (.83)	0 (0)	1 (.73)	1 (.97)
GPT-4o+PST	1 (1)	1 (1)	1 (1)	1 (.76)	1 (.93)	1 (.93)
Llama3.1-70b+PST	0 (0)	1 (.64)	1 (.79)	0 (0)	.82 (.41)	.82 (.82)
Llama3.1-405b+PST	0 (0)	1 (.86)	1 (.86)	1 (.36)	1 (.36)	1 (.73)

QUESTIONS?

